Cybersecurity Research Project – Router Vulnerability Analysis

Project Overview

This research project analyzed the security of a home router web interface in a controlled lab environment to understand real-world web vulnerabilities.

- Date: 10/02/2026 at 22:26
- Issued by: Ahmed MAHINE
- Duration: 2 hours
- Purpose: Educational security assessment

Router Type (target system)

TP-Link TD-W8961N, Firmware Version v5 (Sep 28, 2023), Hardware Version v5

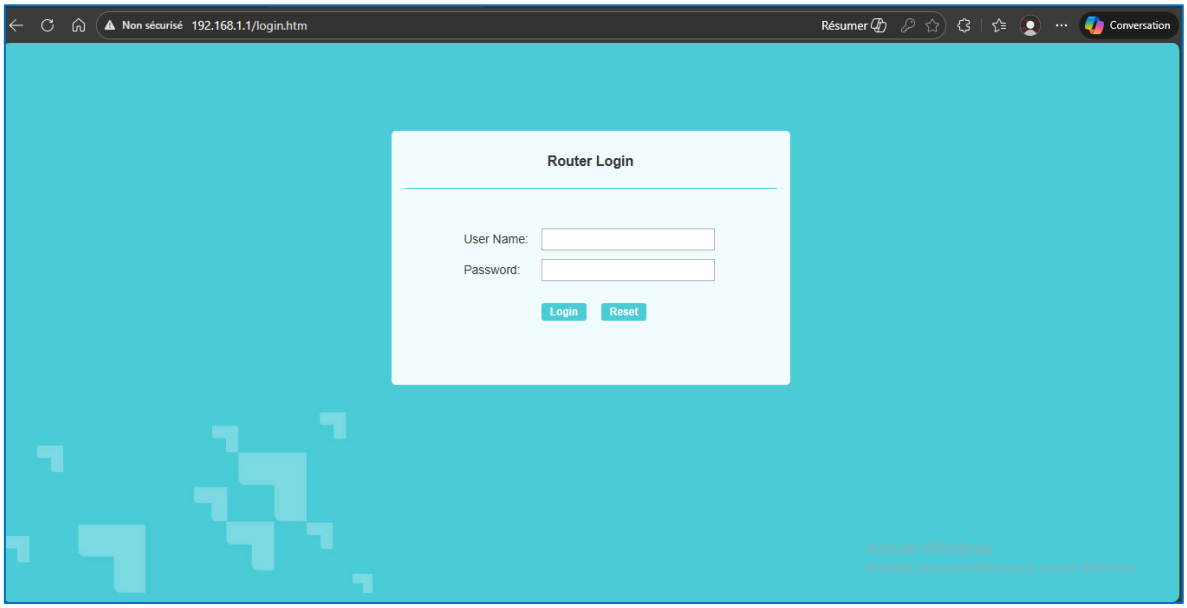

Figure 1 – Router administrative login interface (target system)

Security Analysis Approach

Systematic payload testing, filter behavior analysis, service enumeration, and impact evaluation were conducted following ethical testing practices.

```
Report issues at https://termux.dev/issues
~ $ nmap -sV 192.168.1.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-10 19:15 +0100
Stats: 0:00:55 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 57.14% done; ETC: 19:16 (0:00:39 remaining)
Nmap scan report for 192.168.1.1
Host is up (0.0096s latency).
Not shown: 993 closed tcp ports (conn-refused)
PORT     STATE SERVICE    VERSION
21/tcp   open  ftp?
22/tcp   open  ssh        (protocol 2.0)
23/tcp   open  tcpwrapped
53/tcp   open  domain?
80/tcp   open  http       RTK Web 0.9
443/tcp  open  ssl/https?
5431/tcp open  upnp       MiniUPnP
2 services unrecognized despite returning data. If you know the service/version, please submit th
```

Figure 2 – Service enumeration showing exposed network services on the router

Key Technical Discovery

A reflected cross-site scripting vulnerability was identified in a diagnostic tool due to insufficient input sanitization and inconsistent filtering rules.

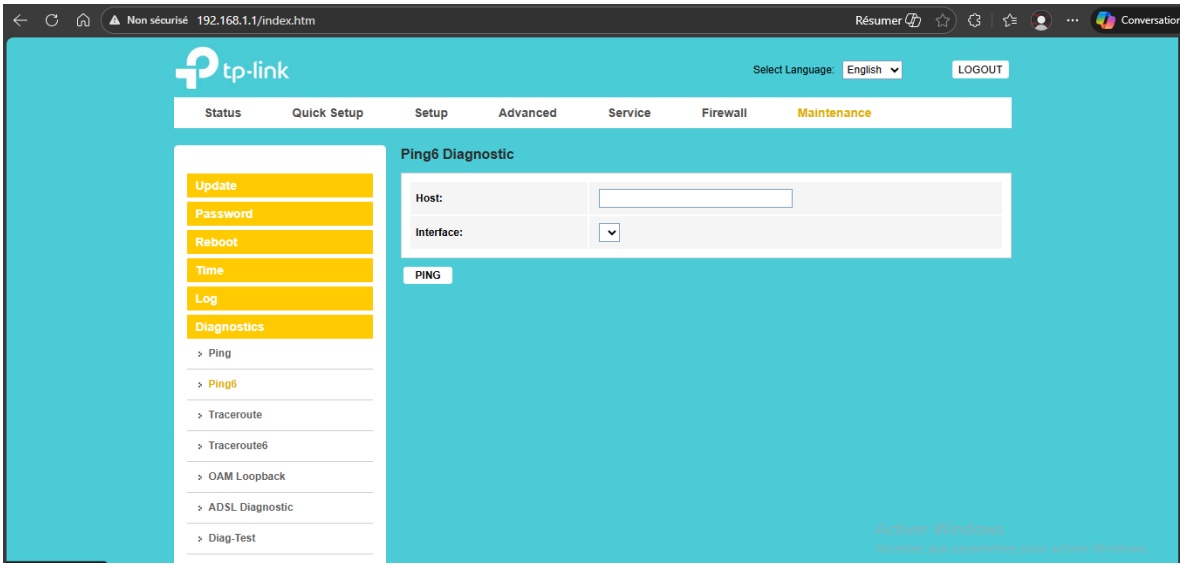

Figure 3 – Diagnostic interface containing the vulnerable input field

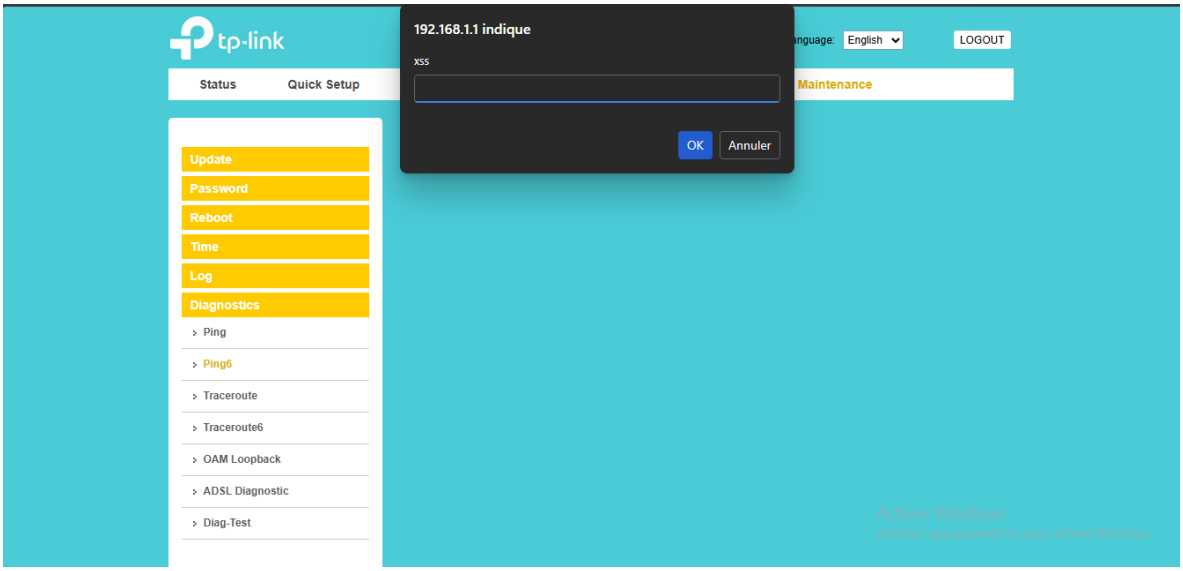

Figure 4 – Proof of client-side script execution during controlled testing.

Skills Demonstrated

- Network scanning and enumeration
- Web vulnerability analysis
- Input validation testing
- Security risk assessment
- Defensive remediation planning

Risk Understanding

The study evaluated how client-side vulnerabilities can lead to phishing, CSRF, and administrative compromise if not mitigated.

```
C:\Users\HP>python3 -m http.server 9999
Serving HTTP on :: port 9999 (http://[::]:9999/) ...
::ffff:192.168.1.6 - - [10/Feb/2026 14:26:23] "GET / HTTP/1.1" 200 -
```

Figure 4 – Proof of client-side script execution during controlled testing.

Defensive Engineering Knowledge

Recommendations include strict input validation, output encoding, Content Security Policy, service hardening, and secure configuration practices.

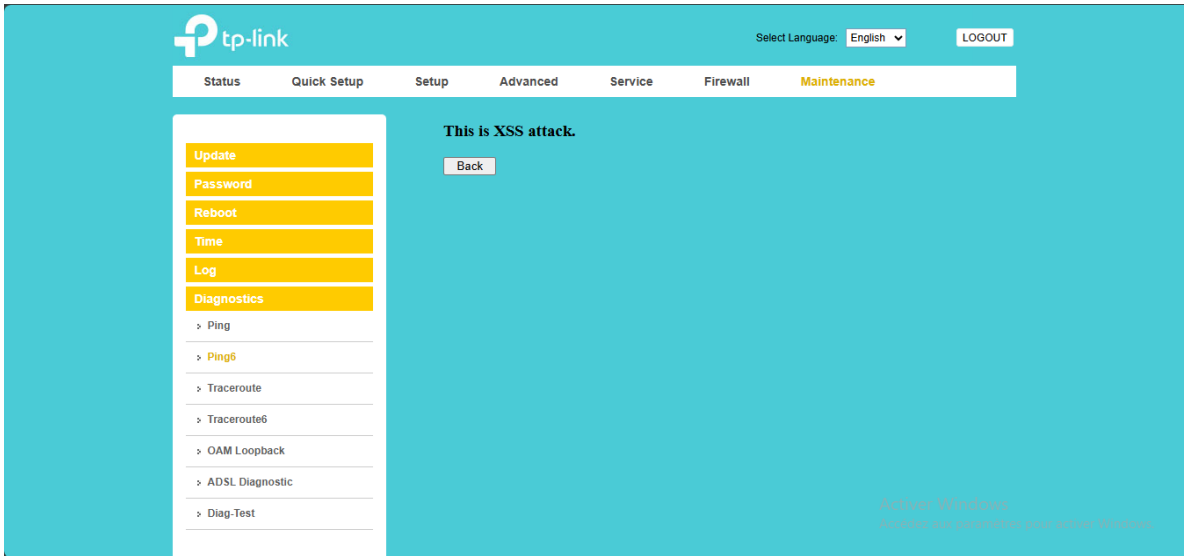

Figure 6 – Built-in filtering mechanism blocking script tag payloads

Router Security Best Practices

- Keep firmware updated
- Use HTTPS-only admin panel
- Disable unused services
- Apply strong authentication
- Restrict management interfaces
- Monitor network logs
- Apply principle of least privilege

Educational Outcome

This project strengthened practical understanding of web security, attack surface analysis, and defensive cybersecurity engineering.

Ethics Statement

All testing was performed in an assessment was performed solely on personally owned equipment